

# VISHAL SHARMA

+91-8218837039 | [sharma2005vishal@gmail.com](mailto:sharma2005vishal@gmail.com) | Kanpur, India

[sharmavishal2.github.io/Portfolio](https://sharmavishal2.github.io/Portfolio) | [linkedin.com/in/Vishal](https://linkedin.com/in/Vishal) | [github.com/sharmaVishal2](https://github.com/sharmaVishal2) | [leetcode.com/Vishal](https://leetcode.com/Vishal)

## PROFESSIONAL SUMMARY

Software Development Engineer / Java Backend Developer with expertise in building scalable microservices, RESTful API architectures, and distributed systems using Spring Boot, Spring Security, Redis, and PostgreSQL. Strong foundation in Data Structures & Algorithms (300+ problems solved), Low-Level Design (LLD), concurrency control, and relational database query optimization. Demonstrated technical leadership by guiding a 6-member engineering team to a Top 50 finish at Smart India Hackathon.

## TECHNICAL SKILLS

**Programming Languages:** Java (Core & Advanced), Python, JavaScript, SQL

**Backend Frameworks & Architecture:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate ORM, REST APIs, Microservices

**Distributed Systems & Databases:** Redis (Distributed Caching, Rate Limiting), PostgreSQL, MySQL, MongoDB, Concurrency & Multithreading

**DevOps, Tools & Verification:** Docker, Git, GitHub, Maven, Postman, JUnit 5, CI/CD Basics, Vercel, Render

**Core Computer Science & Cloud:** OOP, System Design (LLD/HLD), Multithreading, OAuth 2.0, JWT, LLM Integration

## TECHNICAL PROJECTS

### Distributed Rate Limiter Service

*Spring Boot, Redis, PostgreSQL, Docker, Spring Data JPA*

[github.com/sharmaVishal2/Distributed-Rate-Limiter](https://github.com/sharmaVishal2/Distributed-Rate-Limiter)

- Engineered a distributed API rate-limiting middleware to throttle traffic spikes and prevent cascading microservice outages.
- Implemented configurable **Token Bucket** and **Sliding Window Counter** algorithms, maintaining **sub-5ms evaluation latency** under concurrent loads.
- Utilized **Redis atomic operations (INCR, EXPIRE)** to ensure thread-safe distributed state management and eliminate race conditions.
- Containerized multi-node backend services using Docker and automated functional verification with JUnit 5 unit/integration tests.

### E-Commerce Backend Platform

*Spring Boot, PostgreSQL, Spring Security, OAuth 2.0, Hibernate*

[github.com/sharmaVishal2/E-Commerce-Application-by-Java-Spring-Boot](https://github.com/sharmaVishal2/E-Commerce-Application-by-Java-Spring-Boot)

- Architected a modular enterprise backend supporting product catalog, shopping cart state management, order processing, and payment lifecycles.
- Secured endpoints with **Spring Security**, implementing Role-Based Access Control (RBAC), stateless **JWT authentication**, and **OAuth 2.0** social login.
- Optimized complex relational database queries via PostgreSQL composite B-Tree indexing and Hibernate batch fetching, slashing response times by **~15%**.
- Implemented dynamic multi-attribute search and paginated filtering mechanisms using Spring Data JPA Specifications.

## EXPERIENCE & CERTIFICATIONS

- Software Engineering Virtual Experience – JPMorgan Chase & Co.** (Forage, 2026): Interfaced and optimized RESTful backend microservices, streamlined financial data feeds, and validated system reliability using JUnit 5.
- Generative AI Essentials & Cybersecurity Awareness** (MPIT-CoE × TCS Foundation, TCS iON, 2026): Certified in LLM API integration, prompt engineering, secure API endpoints, and vulnerability management.

## HONORS & ACHIEVEMENTS

- Smart India Hackathon (SIH):** Team Leader for 6-member squad; shortlisted among **Top 50 teams nationwide**.
- Competitive Programming:** Solved **300+ DSA problems** across LeetCode and CodeChef (Global Rank **48** in PLACEPREP03; Global Rank **824** in Starter Contest 237).
- National Entrepreneurship Challenge (NEC):** Secured **AIR 102** among 4,000+ participating engineering institutions (IIT Bombay).

## EDUCATION

**Dr. Ambedkar Institute of Technology for Divyangjan (AKTU), Kanpur**

2023 – 2027 (Expected)

Bachelor of Technology in Computer Science and Engineering

*CGPA: 7.04 / 10*

**Relevant Coursework:** System Design (LLD/HLD), Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks

**Kanha Makhan Public School (CBSE), Mathura**

2019 – 2021

Senior Secondary (Class XII) – 74.2% | Secondary (Class X) – 80.4%